Telnet and SSH

Quick Notes, Common Commands and 10 Practice Questions

1. Quick Comparison

Feature	Telnet	SSH
Port	TCP 23	TCP 22
Encryption	None (plain text)	Yes (encrypted)
Authentication	Password sent in clear text	Password or public key
Security	Insecure, avoid in production	Secure, recommended
Use today	Labs, testing a port	Remote admin, file transfer (SCP/SFTP)

2. Cisco IOS: Configure Telnet

```
R1> enable
R1# configure terminal
R1(config)# enable secret cisco123
R1(config)# line vty 0 4
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# transport input telnet
R1(config-line)# end
R1# copy running-config startup-config
```

3. Cisco IOS: Configure SSH

```
R1(config)# hostname R1
R1(config)# ip domain-name aru.local
R1(config)# username admin secret Cisco123
R1(config)# crypto key generate rsa modulus 2048
R1(config)# ip ssh version 2
R1(config)# line vty 0 4
R1(config-line)# login local
R1(config-line)# transport input ssh
R1(config-line)# exit
R1(config)# interface vlan 1
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
```

Note: SSH needs a hostname (not the default), a domain name, RSA keys and a local user. Use `transport input telnet ssh` to allow both.

4. Verification Commands

Command	Purpose
`show ip ssh`	Shows SSH version and status
`show ssh`	Shows active SSH sessions
`show users`	Shows who is connected to the device
`show line vty 0 4`	Shows VTY line settings
`show running-config | section line vty`	Shows only the VTY config

`show crypto key mypubkey rsa`	Shows generated RSA keys
`show sessions`	Shows outgoing Telnet/SSH sessions
`ping 192.168.1.1`	Test connectivity first

5. Client Commands (PC, Linux, Windows, Packet Tracer)

Command	Purpose
`telnet 192.168.1.1`	Telnet to a device (port 23)
`telnet 192.168.1.1 80`	Test if a TCP port is open
`ssh admin@192.168.1.1`	SSH using username@host
`ssh -l admin 192.168.1.1`	SSH using -l for username (Cisco/Packet Tracer style)
`ssh -p 2222 admin@host`	SSH on a custom port
`ssh -v admin@host`	Verbose mode for troubleshooting
`ssh-keygen -t rsa -b 4096`	Generate an SSH key pair
`ssh-copy-id admin@host`	Copy public key to server
`scp file.txt admin@host:/tmp/`	Copy a file over SSH
`sftp admin@host`	Secure file transfer session
`exit or logout`	Close the session
`Ctrl+Shift+6, then x`	Suspend a Cisco session and return to the local router

6. Common Problems

- **Password required, but none set:** the VTY line has `login` but no `password`.
- **Connection refused:** the transport input does not allow that protocol, or the VTY lines are all in use.
- **SSH key error:** no RSA key generated, or hostname/domain name missing.
- **Cannot reach device:** check IP address, interface status, default gateway and ACLs.

7. Practice Questions

Q1. What port numbers do Telnet and SSH use?

Q2. Why is Telnet considered insecure? Explain what an attacker with a packet sniffer can see.

Q3. List the four things required on a Cisco router before SSH can work.

Q4. Write the commands to generate a 2048-bit RSA key and force SSH version 2.

Q5. What does the command `transport input ssh` do on the VTY lines?

Q6. What is the difference between `login` and `login local`?

Q7. Write the commands to allow only SSH on VTY lines 0 to 4 using the local user database, with user *admin* and secret *Cisco123*.

Q8. Which command shows the active SSH sessions? Which shows the SSH version enabled?

Q9. From a Linux terminal, write the command to SSH to 10.0.0.5 as user *student* on port 2222.

Q10. A technician cannot SSH into R1 but can ping it. Give three possible causes and the command you would use to check each.

8. Answer Key

Q1. Telnet uses TCP 23. SSH uses TCP 22.

Q2. Telnet sends everything, including usernames and passwords, in plain text. A sniffer such as Wireshark can read the full session.

Q3. (1) A hostname other than the default, (2) a domain name, (3) RSA keys generated, (4) a local username and password. Also: VTY lines set to login local and transport input ssh.

Q4.
```
R1(config)# crypto key generate rsa modulus 2048
R1(config)# ip ssh version 2
```

Q5. It allows only SSH connections on the VTY lines. Telnet connections are refused.

Q6. `login` uses the single password set on the line. `login local` uses usernames and passwords from the local database (`username ... secret ...`).

Q7.
```
R1(config)# username admin secret Cisco123
R1(config)# line vty 0 4
R1(config-line)# login local
R1(config-line)# transport input ssh
```

Q8. `show ssh` (or `show users`) for active sessions. `show ip ssh` for the SSH version.

Q9. `ssh -p 2222 student@10.0.0.5`

Q10. (a) No RSA key or missing domain name: `show ip ssh` / `show crypto key mypubkey rsa`. (b) VTY blocks SSH: `show running-config | section line vty`. (c) No local user or wrong login method: `show running-config | include username`. Also possible: an ACL on the VTY lines or all VTY lines busy (`show users`).

Tip: Practice these in Packet Tracer: build 1 router + 1 switch + 2 PCs, configure Telnet first, then switch to SSH and compare.